Cultural Matching Analysis

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Overview

This analysis file examines the role of cultural matching and network opacity in predicting the persistence of social ties over time. We pool data across adjacent waves from the NetSense study to construct a person-period dataset, restricting the sample to non-kin, on-campus student ties. We estimate the probability of tie persistence as a function of cultural matching (closed-form domains and open-ended activities) and network opacity (the number of "Don't Know" responses regarding an alter's preferences) using mixed-effects logistic regression (`lme4::glmer`).

Main Effects Models

The following models assess the main effects of closed-form cultural matching, open-ended cultural matching, and network opacity on the odds of tie persistence. Model 1 shows that closed-form matches independently predict persistence. Model 2 introduces open-ended matches, demonstrating the value of incorporating both open and closed-form items. Model 3 incorporates network opacity (the number of "Don't know" responses regarding an alter's preferences). The main effects model reveals that a lack of shared cultural knowledge signals a weaker tie.

Table 1: Odds Ratios for Tie Persistence (Main Effects Models)

	Model 1: Closed	Model 2: + Open	Model 3: + Opacity
Closed-Form Matches	1.225***	1.174**	
	(0.060)	(0.059)	
Open-Ended Matches		1.322***	1.310***
		(0.072)	(0.071)
Network Opacity			0.748***
			(0.042)
Subjective Closeness: Somewhat	0.526***	0.574***	0.583***
	(0.073)	(0.081)	(0.082)
Subjective Closeness: Close	0.332***	0.365***	0.419***
	(0.075)	(0.083)	(0.097)
Tie Duration (Years)	1.026	1.015	0.987
	(0.048)	(0.048)	(0.048)
Tie Duration Squared	0.998	0.999	1.000
	(0.003)	(0.003)	(0.003)
Ego: Woman	1.084	1.045	1.016
	(0.404)	(0.392)	(0.385)
Alter: Woman	1.414*	1.300	1.247
	(0.240)	(0.224)	(0.217)
Ego Woman × Alter Woman	0.622+	0.714	0.750
	(0.163)	(0.190)	(0.202)
Roommate/Dormmate	1.117	1.031	0.886
	(0.441)	(0.413)	(0.357)
Friend	0.811	0.775	0.727
	(0.224)	(0.217)	(0.204)
Frequency: Weekly/Daily	1.120	1.158	1.115
	(0.184)	(0.192)	(0.187)
Race Homophily: Same Race	0.967	0.961	0.921
	(0.191)	(0.191)	(0.185)
Race Homophily: Unknown	0.162***	0.165***	0.157***
	(0.030)	(0.031)	(0.030)
Time Period: Wave 2-5	0.053***	0.074***	0.063***
	(0.018)	(0.025)	(0.022)
Time Period: Wave 5-6	0.090***	0.120***	0.088***
	(0.031)	(0.042)	(0.032)
Num.Obs.	3251	3251	3251
AIC	2562.3	2536.7	2518.5
BIC	2665.8	2646.3	2628.1

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Marginal Effects

Figure Figure 1 visualizes the marginal effect (calculated as the change from the minimum to the maximum value) of closed-form matches, open-ended matches, and network opacity on the predicted probability of tie persistence.

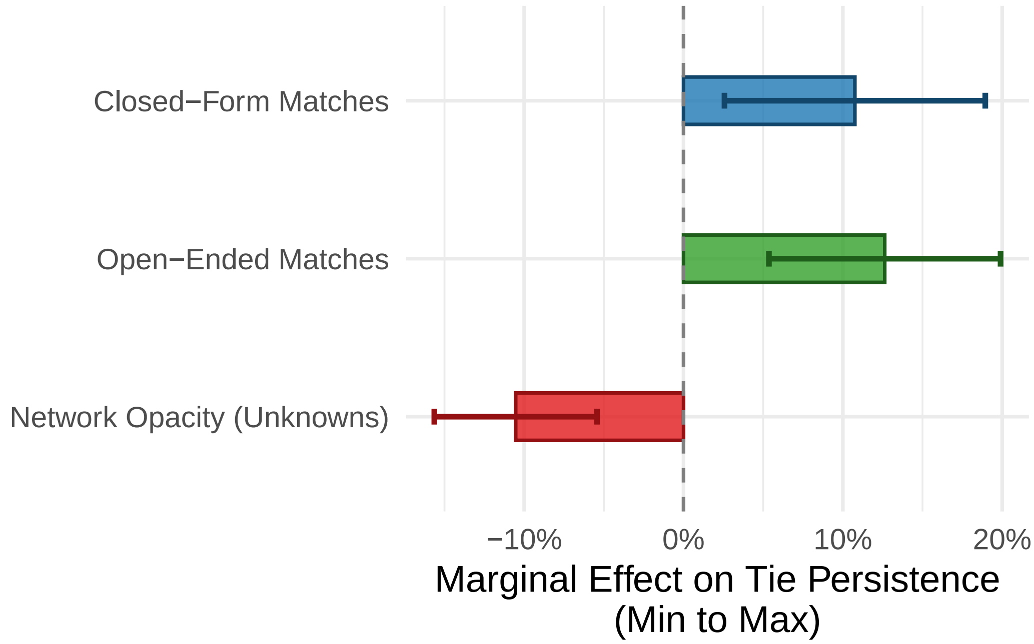

Figure 1: Marginal Effects of Cultural Matches and Network Opacity (Min to Max)

Subjective Closeness Interactions

To test whether the protective effects of cultural matching or the detrimental effects of network opacity depend on the strength of the tie, we estimated auxiliary interaction models between subjective closeness and each of the three cultural variables. While interaction terms in logistic regression models can be difficult to interpret directly from coefficients due to the non-linear link function, computing predicted probabilities across the range of the variables provides a clearer picture of the underlying dynamics.

Figure Figure 2 plots the marginal effects of the three cultural variables across the three levels of subjective closeness.

As the marginal effects plot clearly demonstrates, the role of cultural matching varies substantially by subjective closeness. For "Not Close" ties (the steepest slopes), an increase in shared

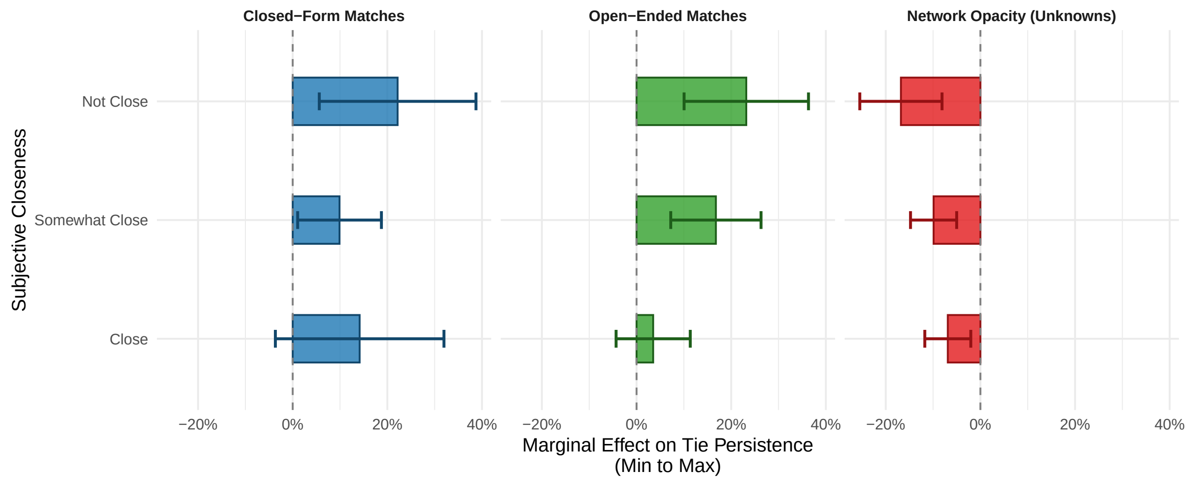

Figure 2: Predicted Probability of Tie Persistence by Cultural Matching, Network Opacity, and Subjective Closeness

cultural tastes—whether measured through closed-form domains or open-ended activities—dramatically increases the probability of tie persistence. Conversely, for ties that are already "Close," the slope is essentially flat. This suggests that shared cultural tastes provide critical scaffolding that helps weak ties survive, whereas strong ties are resilient to decay regardless of cultural matching. Similarly, network opacity exerts its most severe negative effect on ties that are not subjectively close.

Appendix: Descriptive Statistics

Table 2: Descriptive Statistics for Continuous Variables

Variable	Mean	SD	Min	Max
Closed-Form Matches	2.12	1.47	0.00	6
Open-Ended Matches	2.44	1.32	0.00	5
Cultural Network Opacity	0.81	1.51	0.00	6
Tie Duration (Years)	3.01	4.19	0.08	20

Table 3: Descriptive Statistics for Categorical Variables

Variable	Category	N	Percent
Alter Gender	Men	1160	53.5
Alter Gender	Women	1007	46.5
Ego Gender	Men	1127	52.7
Ego Gender	Women	1010	47.3
Friend	No	236	10.9
Friend	Yes	1939	89.1
Race Homophily	Heterophilous	217	10.0
Race Homophily	Homophilous	500	23.0
Race Homophily	Unknown	1458	67.0
Roommate/Dormmate	No	2062	94.8
Roommate/Dormmate	Yes	113	5.2
Subjective Closeness	Close	302	14.1
Subjective Closeness	Not Close	712	33.3
Subjective Closeness	Somewhat Close	1127	52.6
Tie Persisted	No	1489	68.5
Tie Persisted	Yes	686	31.5
Time Period	Wave 1 to 2	104	4.8
Time Period	Wave 2 to 5	1365	62.8
Time Period	Wave 5 to 6	706	32.5